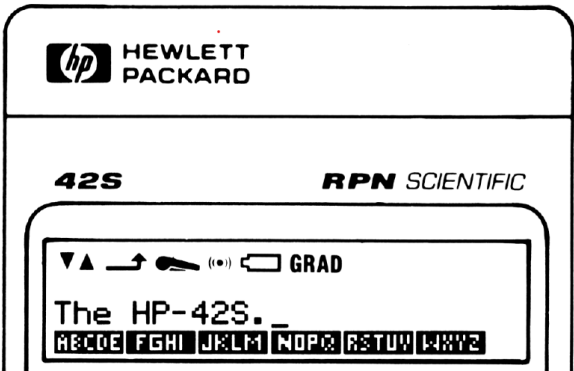


Application note 0026: HP-42S Compatibility



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Revision	Date	Description
1.0	2026-07-10	First publication.

Section 1: Overview

The 47 calculator (C47 or R47, referred to as 47 here) shares many features and functions with the HP-42S (referred to as 42S here), however it departs from its predecessor in many ways which prevent 100% compatibility.

This application note provides some details about the similarities as well as the differences between the 42S and the 47 to help convert HP-42S programs to the 47.

For reference, Appendix A provides the complete list of the 42S functions with their 47 equivalents, or with short notes when there is no equivalent.

Similarly, Appendix B provides the list of the 42S flags and 47 equivalents.

Section 2: Stack

On the 42S the stack size is 4 registers (X, Y, Z & T) and the entry mode is [classic RPN](#) where ENTER pressed to close a number entry duplicates this entry in both X and Y registers, pushing the stack up and disabling stack lift for the entry.

On the 47 the stack depth is configurable¹, either 4 registers (X, Y, Z & T) or 8 registers (X, Y, Z, T, A, B, C & D). By default the stack size is set to 8, which may result that a 42S program relying on a 4-level stack with duplication of the T register may not work with an 8-level stack. It is recommended when running a 42S converted program to set the stack size to 4.

On the 47 the entry mode is also configurable², either classic RPN as the 42S, or entry RPN where ENTER doesn't duplicate the entry in the Y register and doesn't disable stack lift. The entry mode should not have an impact on program execution as an ENTER instruction in a program always lift the stack whatever the entry mode selected.

Section 3: Alpha register

String handling is one of the main differences between the 42S and the 47 and is likely one of the first issues faced when converting a 42S program to 47.

The HP-42S as its predecessor, the HP-41 has a dedicated Alpha register, separated from the stack and the other registers to store strings. The functions displaying or operating on strings implicitly use this Alpha register include **PROMPT**, **AVIEW** and more.

The 47 doesn't have a dedicated Alpha register but is more versatile as any register or variable can be used to store a string. The first consequence is that strings are entered into the X register on the stack as you would enter any other data, and not entered directly into a dedicated register. A string in a 42S program is directly entered in the Alpha register without disturbing the stack, while the same string in a 47 program will be entered into the X register, changing the stack content, which could change program behavior if not accounted for.

¹ Select **SSIZE4** or **SSIZE8** in **f MODE** on the C47 and in **f PREF** on the R47

² Enable or disable **ePRN** (for entry RPN) in **f MODE** on the C47 and in **f PREF** on the R47

The second consequence of using standard strings and string functions in 47 is that functions handling strings need to specify explicitly where the string would be stored. For example where **AVIEW** on the 42S doesn't need any parameter to view the content of the Alpha register, **AVIEW** on the 47 needs a parameter to specify where the string is located, eg: **AVIEW Z**, **AVIEW 12** or **AVIEW 'Result'** to respectively display the string stored in register Z, 12 or the variable 'Result'.

Converting string instructions in a 42S program to 47 native string instructions requires an analysis of the program to understand how it works, where to store the strings, and how to manage the impact on the stack. To streamline this conversion process with automated tools such as rejig, a dedicated set of 42S compatible string instructions has been added to the 47. These instructions do not replace the native 47 string instructions, but rather provide an easy path to convert 42S programs.

The 42S-compatible dedicated instructions start with 42 and are available in the **42** menu, accessible with **f CAT MENUS**. The first page contains the functions related to strings. All these functions use register K as the implicit Alpha register.

42KEYG	42KEYX	42VRMNU	42PRA	42ATOX	42XTOA
42ALENG	42POSA	42AROT	42ASHF	42PRMPT	42AIP
42α	42┌	42ARCL	42ASTO	42AVIEW	42CLA

The first two functions are for entering strings:

- **42α 'string'** stores the string in the register K, it replaces "**string**" in a 42S program.
- **42┌ 'string'** appends the string to the one in the register K, it replace **┌**"**string**" in a 42S program.

The maximum number of characters accepted is the same as on the HP-42S and this originated from the HP-41 internal string coding (15 for **42α** and 14 for **42┌**).

The other functions in the menu above are direct equivalents of their 42S counterparts:

- **42ARCL r** appends the content of the specified register r to the K register
- **42ASTO r** copies the first six characters in the K register into the specified register r
- **42AVIEW** view the string in register K
- **42CLA** stores an empty string in register K
- **42ALENG** returns the length of the string in the register K in X, and returns 0 for an empty string
- **42POSA** searches the K register for the target specified in X. If found, returns the character position starting from 0; if not found, returns -1
- **42AROT** rotates the K register by the number of characters specified in X. The sign of X determines direction of rotation: positive to the left; negative to the right
- **42ASHF** shifts the six left-most characters out of the K register
- **42PRMPT** displays the K register and halts program execution
- **42AIP** appends the integer part of X to the K register
- **42KEYG** programmable menu: on menu key, go to. Defines the label to be branched to when a menu key is pressed, using the string in the register K as the key label (note the 47 native **KEYG** takes the string from the X register)
- **42KEYX** programmable menu: on menu key, execute. Defines the label to be executed as a subroutine when the menu key is pressed, using the string in the register K as the key label (note the 47 native **KEYX** takes the string from the X register)
- **42VRMNU** creates a variable menu using **MVAR** instructions following the specified global label. The label of a key pressed without data input is stored in the K register. (The 47 native **VARMENU** also stores it in

- the K register but a dedicated 42 instruction ensures that a converted program will continue to work correctly, should in future the 47 Alpha register gets assigned to another register
- **42PRA** prints the content of the K register as an alpha string
- **42ATOX** converts the left-most character in the K register to its character code (returned to the X-register and deletes the character from the Alpha register
- **42XTOA** appends a character (specified by the character code in the X-register) to the Alpha register. If the X-register contains an Alpha string, appends the entire string. If the X-register contains a matrix, the **42XTOA** function uses each element in the matrix as a character code

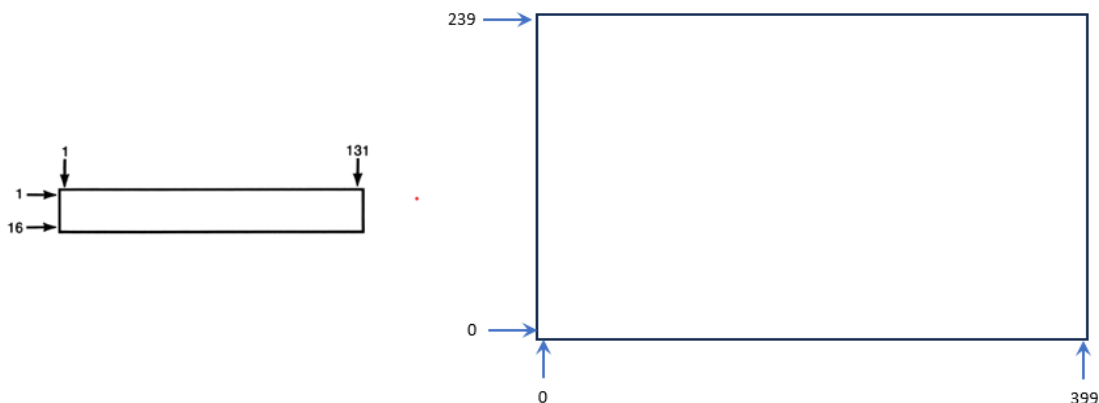
To ensure that the register K follows the HP-42S 44-character limit, the maximum string size is enforced when executing the functions **42PRA**, **42AIP**, **42ARCL** and **42XTOA**, as they can increase the length of the string in the K register.

AON and **AOFF** currently don't have any direct 47 equivalent. The closest to **AON** is **SF ALPHA** which will set the Alpha Input mode but the string entered will be placed in register X and not in register K, and it's not possible to press **R/S** in AIM to continue the program execution as **R/S** is entering the character '?' in AIM on the 47.

Section 4: Graphic functions

The graphic functions are another area where there the differences between the 42S and the 47 will prevent 42S programs such as the Display Plot Program ("DPLOT"), the Printer Plot Program ("PLOT") from the HP-42S manual or programs using custom characters such as the Circuit program ("CIRCT") from the HP-42S Electrical Engineering Step-by-Step solutions manual to run as-is on the 47.

The first difference is the size of the display: 131x16 pixels for the 42S and 400x240 for the 47. The origin point is also different: (1,1) in the top left corner for the 42S and (0,0) in the bottom left corner for the 47 which could lead to 42S graphics being upside down on the 47.



The second difference is how the graphic commands work.

The HP-42S has three commands to draw graphics on its LCD screen:

- **CLLCD** Blanks the entire display (131x16 pixels)
- **PIXEL** Turns on a single pixel (dot) in the display. The location of the pixel is given by the numbers in the X and Y-registers. If values in X and/or Y are negative, PIXEL draws one or two lines vertically ($x < 0$) or horizontally ($y < 0$)
- **AGRAPH** Displays a graphics image. Each character in the Alpha register specifies an 8-dot column pattern. The X- and Y-registers specify the pixel location of the image. The status of flags 34 and 35 determine how the graphics image is displayed

and one command to print the display to an HP-82240A/B IR printer:

- **PRLCD** Prints the entire display

The 47 has similar commands but they don't all work in a similar way:

- **CLLCD** Blanks the entire display (400x240 pixels)³
- **PIXEL** Turns on a single pixel (dot) in the display. The location of the pixel is given by the numbers in the X and Y-registers. If values in X and/or Y are negative, PIXEL draws one or two lines vertically ($x < 0$) or horizontally ($y < 0$). To draw a line at $x=0$ or $y=0$, use either the real value -0. or the LI -400
- **POINT** Similar to PIXEL but displays 9 (3x3) pixels around (X, Y)
- **AGRAPH** Displays a 64 column pattern based on a short integer value in the specified register. The value of the variable GRAMOD determine how the graphics image is displayed

and the command to print the display to an HP-82240A/B IR printer is currently not implemented due to the difference between the display pixel width (400) and the printer dot width (166):

- **PLCD** Prints the entire display – NOT IMPLEMENTED

Section 5: Matrix functions

The matrix functions of the 47 includes the 42S functions, but with a significant difference on the 47: several programmable 42S matrix editing functions are not programmable on the 47:

- **DELR** Delete row. Deletes the current row from the indexed matrix
- **EDIT** Edit a matrix in the X-register
- **EDITN** Edit a named matrix
- **INSR** Inserts a row in the indexed matrix
- **OLD** Recalls the current element from the indexed matrix. (Equivalent to **RCLEL**)
- **←** Moves left one element in the indexed matrix
- **↑** Moves up one element in the indexed matrix
- **↓** Moves down one element in the indexed matrix
- **→** Moves right one element in the indexed matrix

Another difference is related to the manual matrix data input and stack usage:

- on the 42S the stack is available in Matrix Edit mode, so it can be used for any kind of operation before inserting the calculated value in the matrix.
- on the 47 the stack is not available during Matrix Edit mode, and only monadic functions are allowed (such as x^2 , **LOG**, **SIN** ...)

³ The previous **CLLCD** command from the WP43 has been renamed **CLLCDxy** as it clears a part of display based on the values in the X and Y-registers

Section 6: Data type related functions

The HP-42S supports 4 different data types:

- Real numbers
- Complex numbers
- Alpha strings
- Matrices

The 47 supports 10 different data types:

- Long integer numbers
- Real numbers
- Complex numbers
- Times
- Dates
- Alpha strings
- Real matrices
- Complex matrices
- Short integers
- Config

Due to the difference in data types some functions do not exist or do not work the same way.

BASE operations

The HP-42S supports base operations for numbers in four different bases (Binary, Octal, Decimal and Hexadecimal) via the selected Base mode in the BASE menu, but numbers in these base modes don't have a specific data type so there are specific arithmetic operations for numbers in base mode:

- **BASE-** Base subtraction. Returns the 36-bit difference of $y - x$
- **BASE+** Base addition. Returns the 36-bit sum of $y + x$
- **BASE+/-** Base change sign. Returns the 36-bit 2's complement of x
- **BASE÷** Base division. Returns the 36-bit quotient of $y \div x$
- **BASEx** Base multiplication. Returns the 36-bit product of $y * x$

The 47 has a dedicated data type for such numbers: the short integer data type. There is no dedicated base arithmetic operations and the standard arithmetic operators are used with short integers (+, -, CHS, ÷, x).

Note 1: on the 42S, in base mode, the word width is fixed to 36 bits and there is only one sign mode (2's complement), while on the 47 the word size can be selected from 1 to 64 bits and the sign mode can be selected between unsigned, 1's complement, 2's complement and mantissa sign. If the 47 doesn't use settings similar to that of the 42S when doing base operations, the results can be different.

Note 2: **AND**, **OR**, **XOR**, **NOT** are bit-wise operations on the 42S where on the 47 they are bit-wise operations when working with short integers, and logical operations when working on real numbers.

For example: On 42S: **255 ENTER 65 AND** will return **65** (bit-wise **AND**), and on 47: **255 ENTER 65 AND** will return **1** (logical **AND**). To get the result of a bit-wise **AND** on the 47, the numbers need to be entered or converted to short integers: **255₁₀ ENTER 65₁₀ AND** will return **65₁₀**.

Note 3: the functions **→DEC** and **→OCT** are included on the HP-42S to provide program compatibility with the HP-41 and are neither related to the 42S BASE application nor have equivalents on the 47.

Hours-minutes-seconds format

The HP-42S has four functions for working with numbers expressed in hours-minutes-seconds format (H.MMSSss), either time values or angles expressed in degrees (D.MMSSss).

- **→HMS** To hours, minutes, and seconds. Converts x hours from a decimal fraction to the minutes-seconds format.
- **→HR** To hours. Converts x from a minutes-seconds format to a decimal fraction.
- **HMS-** Subtracts x from y using H.MMSSss format
- **HMS+** Adds x and y using H.MMSSss (hours-minutes-seconds) format

The 47 supports the first 2 conversion functions:

- **HR→HMS** To hours, minutes, and seconds. Converts x from a decimal fraction to a minutes-seconds format (sexagesimal).
- **HMS→HR** To hours. Converts x from a minutes-seconds format (sexagesimal) to a decimal fraction.

The last 2 listed 42S functions (**HMS-**, **HMS+**) for arithmetic with Minutes and Seconds is done on 47 with the standard **+** and **-** operators on numbers in the Time data type. To convert a number to and from a Time data type (to/from an HMS formatted real), use the **H.MS→T** and **.ms^-1** conversion functions.

For more details about the 47 sexagesimal functions see the app. note 14: [h.ms, d.ms and .ms](#)

Section 7: 42S commands that work differently on the 47

Some 42S commands exist on the 47 but work differently, either with different parameters or with different output.

- **AGRAPH** Alpha graphics – (see Section 4: Graphic functions)
- **ASSIGN** Not programmable and no CUSTOM menu on the 47. **ASSIGN** on the 47 is used to assign functions, programs, menus or variables to keyboard keys or configurable menus soft keys.
- **BIT?** Tests the x^{th} bit of y. The 47 equivalent is **BS? n**: it tests the n^{th} bit of x (n is 0 based and can be in the range 0 - WSIZE)
- **DELAY** Sets the print delay time to x seconds. The 47 equivalent is **▢DELAY n** where n is the delay time in ticks or 10^{th} of second: **▢DELAY 12** is the equivalent of **1.2 DELAY**, setting the delay time to 1.2s
- **GETKEY** The calculator waits for you to press a key. On the 47 **KEY? r** doesn't wait, it acts as a test, executing the next step if no key has been pressed or skipping it if a key has been pressed and returning the keycode in the specified register. Note also that the 47 keycodes (11 to 85) are different from the 42S keycodes (from 1 to 37)
- **ROTX** Rotates the 36-bit number in the Y-register by x bits. There is no direct equivalent: **RL**, **RR**, **RLn** or **RRn** can be used to rotate the content of the X-register but there is no single command similar to **ROTX**

Section 8: 42S commands not relevant on the 47

Some 42S commands are not relevant on the 47 due to the difference in the calculator implementation and feature set.

Global and Statistic registers

The HP-42S has a variable number of global registers (from 0 to 99, default 25) stored in the **REGS** matrix but on the 47 all 99 global registers are always available and there is no **REGS** matrix.

The HP-42S uses a block of global registers to save the summation coefficients. The current statistical mode determines how many coefficients are saved. In linear mode, registers R11-R16 are used, in **ALLΣ** mode R11-R23 are used. The 47 has 28 summation registers which are independent from the global registers and are always allocated.

The 42S functions related to the management of the global and statistic registers are not relevant on the 47: **SIZE**, **ALLΣ**, **LINΣ**, **ΣREG** and **ΣREG?**

CUSTOM menu and local labels

The 42S CUSTOM menu has two modes: a key assignments mode and a local labels mode where it can be used to execute local labels A-J and a-e. This provides compatibility with HP-41 programs local labels direct execution.

The 47 provides several user configurable menus (**MyMenu**, **HOME**, **P.FN**, **Myα**) as well as the possibility to create new user menus, but there is no local labels menu mode, and it's not possible in a program to create a user menu or assign items to keys or user configurable menus.

The 42S CUSTOM menu related functions are not relevant on the 47: **CLKEYS**, **KEYASN**, and **LCLLBL**. There is also no equivalent to flags 27 (used to display the CUSTOM menu) and 72 (toggle between the key-assignments and the local labels modes).

Section 9: Flags

The HP-42S uses 100 flags (numbered 00-99) to keep track of various modes, settings, and conditions. They are split in:

- 30 user flags (00-10 and 81-99) – freely usable by user programs.
- 25 control flags (11-35) used to represent certain operating conditions. They can be set or cleared by the user.
- 45 system flags (36-80) They cannot be directly changed by the user but they can be checked.

The 47 has 100 global flags (00-99) but they are all user flags, the 47 system flags are not numbered but named and accessible in the **SYS.FL TAM** menu for the different flags functions.

Several of the HP-42S control and system flags have direct 47 system flag equivalents. Some 42S flags used to encode values are not accessible as flags on the 47 but such values are accessible via set and/or get commands. Finally, some HP-42S flags are not available or don't have an equivalent on the 47.

See Appendix B for the list of the HP-42S flags and their 47 equivalents.

Appendix A: HP-42S to C47/R47 command mapping

HP-42S	Description	C47/R47	Comment
"string"	Place an Alpha string into the Alpha register.	42α 'string'	Place an Alpha string into the K register
└"string"	Append an Alpha string to the current content of the Alpha register.	42└ 'string'	Append an Alpha string to the current content of the K register
ABS	Absolute value.	x	
ACOS	Arc cosine. Returns $\cos^{-1} x$.	ARCCOS	
ACOSH	Arc hyperbolic cosine. Returns $\cosh^{-1} x$.	arcosh	
ADV	Advances the printer paper one line.	ADV	
AGRAPH	Alpha graphics. Displays a graphics image. Each character in the Alpha register specifies an 8-dot column pattern. The X- and Y-registers specify the pixel location of the image.	AGRAPH	Doesn't work as on the 42S. Uses a 64-bit pattern.
AIP	Appends integer part of x to the Alpha register.	42AIP	Same as AIP, using K as the Alpha register
ALENG	Alpha length. Returns the number of characters in the Alpha register.	42ALENG	Same as ALENG, using K as the Alpha register
ALL	Selects the All display format.	ALL	
ALLΣ	Selects ALLΣ (All-statistics) mode, which uses 13 summation coefficients.	N.A.	
AND	Logical AND. Returns x AND y.	AND	
AOFF	Alpha off. Exits from the ALPHA menu.		Missing
AON	Alpha on. Selects the ALPHA menu.		Missing
ARCL	Alpha recall. Copies data into the Alpha register appending it to the current contents. Numbers are formatted using the current display format.	42ARCL	Same as ARCL, using K as the Alpha register
AROT	Alpha rotate. Rotates the Alpha register by the number of characters specified in the X-register.	42AROT	Same as AROT, using K as the Alpha register
ASHF	Alpha shift. Shifts the six left-most characters out of the Alpha register.	42ASHF	Same as ASHF, using K as the Alpha register
ASIN	Arc sine. Returns $\sin^{-1} x$.	ARCSIN	
ASINH	Arc hyperbolic sine. Returns $\sinh^{-1} x$.	arsinh	
ASSIGN	Assigns a function, program, or variable to a menu key in the CUSTOM menu.	ASSIGN	Doesn't work as on the 42S, not programmable on the 47.
ASTO	Alpha store. Copies the first six characters in the Alpha register into a register or variable.	42ASTO	Same as ASTO, using K as the Alpha register
ATAN	Arc tangent. Returns $\tan^{-1} x$.	ARCTAN	
ATANH	Arc hyperbolic tangent. Returns $\tanh^{-1} x$.	artanh	
ATOX	Alpha to X. Converts the left-most character in the Alpha register to its character code (returned to the X-register) and deletes the character.	42ATOX	Same as ATOX, using K as the Alpha register
AVIEW	Alpha view. Displays the Alpha register.	42AVIEW	Same as AVIEW, using K as the Alpha register
BASE-	Base subtraction. Returns the 36-bit difference of y - x.	-	Standard subtraction is used with short integers
BASE+	Base addition. Returns the 36-bit sum of y + x.	+	Standard addition is used with short integers
BASE+/-	Base change sign. Returns the 36-bit 2's complement of x.	CHS	Standard CHS is used with short integers
BASE÷	Base division. Returns the 36-bit quotient of y ÷ x.	÷	Standard division is used with short integers
BASEx	Base multiplication. Returns the 36-bit product of y * x.	x	Standard multiplication is used with short integers
BEEP	Sounds a sequence of four tones.	BEEP	
BEST	Selects the best curve-fitting model for the current statistical data.	BESTF	
BINM	Selects Binary mode (base 2).	BIN	
BIT?	Tests the x^{th} bit of y. If the bit is set, executes the next program line; if the bit is clear, skips the next program line.	BS? n	Tests the n^{th} bit of x (n is 0-based)

HP-42S	Description	C47/R47	Comment
CF	Clears flag nn (00 < nn < 35; 81 < nn < 99).	CF	(See the Flags mapping table)
CLA	Clears Alpha register. If Alpha mode is on and character entry is terminated (no cursor displayed), then $\leftarrow$ also executes the CLA function.	42CLA	Same as CLA, using K as the Alpha register
CLD	Clear display. Clears a message from the display.	CLD	
CLKEYS	Clears all CUSTOM menu assignments.		No direct equivalent
CLLCD	Clear LCD (liquid crystal display). Blanks the entire display.	CLLCD	
CLMENU	Clear MENU. Deletes all menu key definitions for the programmable menu.	CLMENU	
CLP	Clears a program from memory.	DELP	Not Programmable
CLRG	Clears all of the numbered storage registers to zero.	CLREGS	
CLST	Clears the stack registers to zero.	CLSTK	
CLV	Clears a variable from memory.	DELITM	
CLX	Clears X-register to zero. If digit entry is terminated (no cursor in the display), $\leftarrow$ also executes CLX.	CLx	
CLΣ	Clear statistics. Clears the accumulated statistical data in the summation registers.	CLΣ	
COMB	Combinations of y items taken x at a time. Returns $y! \div (x!(y-x)!)$.	Cyx	
COMPLEX	Converts two real numbers (or matrices) into a complex number (or matrix). Converts a complex number (or matrix) into two real numbers (or matrices).	COMPLEX	
CORR	Returns a correlation coefficient using the current statistical data and curve-fitting model.	CORR	
COS	Cosine. Returns cos x.	COS	
COSH	Hyperbolic cosine. Returns cosh x.	cosh	
CPX?	If the X-register contains a complex number, executes the next program line; if the X-register does not contain a complex number, skips the next program line.	CP?	
CPXRES	Complex-results. Enables the calculator to return a complex result, even if the inputs are real numbers.	SF CPXRES	
CROSS	Returns the cross product of two vectors (matrices or complex numbers).	CROSS	
DECM	Selects Decimal mode (base 10).	DEC	
DEG	Selects the Degrees angular mode.	DEG	
DEL	Deletes the specified number of lines from the current program. Program-entry mode must be on. (Not programmable.)		Missing
DELAY	Sets the print delay time to x seconds.	DELAY	Different argument
DELR	Delete row. Deletes the current row from the indexed matrix.	M.DELR	Not programmable
DET	Returns the determinant of the matrix in the X-register.	M	
DIM	Dimensions a matrix to x columns and y rows. If the matrix does not exist, DIM creates it.	M.DIM	
DIM?	Returns the dimensions of the matrix in the X-register (rows to the Y-register and columns to the X-register).	42DIM#	
DOT	Returns the dot product of two vectors (matrices or complex numbers).	DOT	
DSE	Decrement, Skip if (less than or) Equal. Given ccccccc.fffii in a variable or register, decrements ccececec by ii and skips the next program line if ccececcc is now < fff.	DSE	
EDIT	Edit a matrix in the X-register.	M.EDIT	Not programmable
EDITN	Edit a named matrix.	M.EDITN	Not programmable
END	End of a program.	END	

HP-42S	Description	C47/R47	Comment
ENG	Selects Engineering display format.	ENG	
ENTER	Separates two numbers keyed in sequentially; copies x into the Y-register, y into the Z-register, and z into the T-register, and loses t.	ENTER	
$E \uparrow X$	Natural exponential. Returns e^x .	e^x	
$E \uparrow X-1$	Natural exponential for values of x which are close to zero. Returns e^x-1 , which provides a much higher accuracy in the fractional part of the result.	e^x-1	
EXITALL	Exits all menus.	EXITALL	
EXPF	Selects the exponential curve-fitting model.	ExpF	
FC?	If the specified flag is clear, executes the next program line; if the flag is set, skips the next program line.	FC?	
FC?C	If the specified flag is clear, executes the next program line; if the flag is set, skips the next program line. Cleared after the test is complete. (This function can be used only with flags 00 through 35 and 81 through 99.)	FC?C	
FCSTX	Forecasts an x-value given a y-value.	$\hat{x}$	
FCSTY	Forecasts a y-value given an x-value.	$\hat{y}$	
FIX	Selects Fixed-decimal display format.	FIX	
FNRM	Returns the Frobenius norm of the matrix in the Xregister.	ENORM	
FP	Returns the fractional part of x.	FP	
FS?	If the specified flag is set, executes the next program line; if the flag is clear, skips the next program line.	FS?	
FS?C	If the specified flag is set, executes the next program line; if the flag is clear, skips the next program line. Clears the flag after the test is complete. (This function can be used only with flags 00 through 35 and 81 through 99.)	FS?C	
GAMMA	Gamma function. Returns $\Gamma(x)$.	$\Gamma(x)$	
GETKEY	Get key. The calculator waits for you to press a key. When you do, the key number is returned to the X-register. Keys are numbered from 1 through 37 for normal keys and 38 through 74 for shifted keys.	KEY?	Doesn't work as on the 42S
GETM	Get matrix. Copies a submatrix into the X-register from the indexed matrix.	M.GET	
GRAD	Selects Grads angular mode.	GRAD	
GROW	Selects Grow mode. Executing $\rightarrow$ or J+ causes the matrix to grow by one new row if the index pointers are at the last (lower-right) element in the matrix.	SF M.GROW	
GTO	Go to label. From the keyboard, moves the program pointer to the specified label. In a running program, causes the program to branch to the specified label.	GTO	
HEXM	Selects Hexadecimal mode (base 16).	HEX	
HMS-	Subtracts x from y using H.MMSSss format.		Missing
HMS+	Adds x and y using H.MMSSss (hours-minutes-seconds) format.		Missing
I-	Decrements the row pointer in the indexed matrix.	I-	
I+	Increments the row pointer in the indexed matrix.	I+	
INDEX	Indexes a named matrix.	INDEX	
INPUT	Recalls a register or variable to the X-register, displays the name of the register or variable along with the contents of the X-register, and halts program execution; pressing [R/S]	INPUT	

HP-42S	Description	C47/R47	Comment
	(or shift [SST]) stores x into the register or variable; pressing [EXIT] cancels. (Used only in programs).		
INSR	Inserts a row in the indexed matrix.	M.INSR	Not programmable
INTEG	Integrates the selected integration program with respect to the specified variable.	$\int$	
INVRT	Returns the inverse of the matrix in the X-register.	[M]-1	
IP	Returns the integer part of x.	IP	
ISG	Increment, Skip if Greater. Given cccccc.fffii in a variable or register, increments cccccc by ii and skips the next program line if cccccc is now > fff.	ISG	
J-	Decrements the column pointer in the indexed matrix.	J-	
J+	Increments the column pointer in the indexed matrix.	J+	
KEYASN	Selects Key-assignments mode for the CUSTOM menu.	N.A.	
KEYG	On menu key, go to. Defines the label to be branched to when a particular menu key is pressed.	42KEYG	Same as KEYG, getting the key label from K as the Alpha register
KEYX	On menu key, execute. Defines the label to be executed (as a subroutine) when a particular menu key is pressed.	42KEYX	Same as KEYX, getting the key label from K as the Alpha register
LASTX	Last x. Recalls the last value of x used in a calculation.	LASTx	
LBL	Label. Identifies programs and routines for execution and branching.	LBL	
LCLBL	Selects Local label mode for the CUSTOM menu.	N.A.	
LINF	Selects the linear curve-fitting model.	LinF	
LINZ	Selects Linear statistics mode, which uses six summation coefficients.	N.A.	
LIST	Prints a portion of a program listing. (Not programmable.)	 LIST	
LOG	Common logarithm. Returns log ₁₀ X.	LOG	
LOGF	Selects the logarithmic curve-fitting model.	LogF	
MAN	Selects Manual print mode.	MAN	
MAT?	If the X-register contains a matrix, executes the next program line; if the X-register does not contain a matrix, skips the next program line.	MATR?	
MEAN	Mean. Returns the mean of x-values ($\Sigma x \div n$) and the mean of y-values ($\Sigma y \div n$).	$\bar{x}$	
MENU	Selects the programmable menu.	MENU	
MOD	Modulo. Returns the remainder for $y \div x$.	MOD	
MVAR	Declares a menu variable.	MVAR	
N!	Factorial. Returns x!.	x!	
NEWMAT	New matrix. Creates a y x x matrix in the X-register.	M.NEW	
NORM	Selects Normal print mode, which prints a record of keystrokes.	NORM	
NOT	Logical NOT. Returns NOT x.	NOT	
OCTM	Selects Octal mode (base 8).	OCT	
OFF	Turns the calculator off (programmable). (Pressing [SHIFT] OFF does not execute the programmable OFF function.)	OFF	
OLD	Recalls the current element from the indexed matrix. (Equivalent to RCLEL.)	M.OLD	Not programmable
ON	Continuous on. Prevents the calculator from automatically turning off after ten minutes of inactivity.	CF AUTOFF	
OR	Logical OR. Returns x OR y.	OR	
PERM	Permutations of y items taken x at a time. Returns $y! = (y - x)!$.	Pyx	
PGMINT	Selects a program to integrate.	PGMINT	

HP-42S	Description	C47/R47	Comment
PGMSLV	Selects a program to solve.	PGMSLV	
PI	Recalls an approximation of π into the X-register (3.14159265359).	π	
PIXEL	Turns on a single pixel (dot) in the display. The location of the pixel is given by the numbers in the X and Y-registers.	PIXEL	
POLAR	Selects Polar coordinate mode for displaying complex numbers.	POLAR	
POSA	Position in Alpha. Searches the Alpha register for the target specified in the X-register. If found, returns the character position; if not found, returns -1.	42POSA	Same as POSA, using K as the Alpha register
PRA	Print Alpha register.	42PRA	Same as PRA, using K as the Alpha register
PRLCD	Print LCD (liquid crystal display). Prints the entire display.	 LCD	
PROFF	Printing off. Clears flags 21 and 55.	 OFF	
PROMPT	Displays the Alpha register and halts program execution.	42PRMPT	Same as PROMPT, using K as the Alpha register
PRON	Printing on. Sets flags 21 and 55.	 ON	
PRP	Print program. If a label is not specified, prints the current program. (Not programmable.)	 PROG	
PRSTK	Print stack. Prints the contents of the stack registers (X, Y, Z, and T).	 STK	
PRUSR	Prints user variables and programs.	 USER	
PRV	Print variable.	 r r	
PRX	Print X-register.	 x	
PRΣ	Print statistics. Prints the contents of the summation registers.	 Σ	
PSE	Pauses program execution for about 1 second.	PAUSE 10	
PUTM	Put matrix. Stores the matrix in the X-register into the indexed matrix beginning at the current element.	M.PUT	
PWRF	Selects the power curve-fitting model.	PowerF	
QUIET	Toggles flag 26 to disable/enable the beeper. (Not programmable.)	FF QUIET	Flip flag QUIET
R ↑	Rolls the contents of the four stack registers up one position.	R↑	
R ↓	Rolls the contents of the four stack registers down one position.	R↓	
R/S	Run/stop. Runs a program (beginning at the current program line) or stops a running program. In Program-entry mode, inserts a STOP instruction into the program.	R/S	
R<>R	Row swap row. Swaps the elements in rows x and y in the indexed matrix.	M.R↔R	
RAD	Selects Radians angular mode.	RAD	
RAN	Returns a random number ($0 < x < 1$).	RAN#	
RCL	Recalls data into the X-register.	RCL	
RCL-	Recall subtraction. Recalls data and subtracts it from the contents of the X-register.	RCL-	
RCL+	Recall addition. Recalls data and adds it to the contents of the X-register.	RCL+	
RCL÷	Recall division. Recalls data and divides it into the contents of the X-register.	RCL÷	
RCLEL	Recall element. Recalls the current matrix element from the indexed matrix.	RCLEL	
RCLIJ	Recalls the row- and column-pointer values (I and J) for the	RCLIJ	

HP-42S	Description	C47/R47	Comment
	indexed matrix.		
RCLx	Recall multiplication. Recalls data and multiplies it by the contents of the X-register.	RCLx	
RDX,	Selects a comma to be used as the radix mark (decimal point).	RCOM,	
RDX.	Selects a period to be used as the radix mark (decimal point).	RPER.	
REAL?	If the X-register contains a real number, executes the next program line; if the X-register does not contain a real number, skips the next program line.	IR?	
REALRES	Real-results. Disables the calculator's ability to return a complex result using real-number inputs.	CF CPXRES	
RECT	Selects Rectangular coordinate mode for displaying complex numbers.	RECT	
RND	Rounds the number in the X-register using the current display format.	ROUND	
RNRM	Returns the row norm of the matrix in the X-register.	RNORM	
ROTXy	Rotates the 36-bit number in the Y-register by x bits.		No direct equivalent
RSUM	Returns the row sum of each row of the matrix in the X-register and returns the sums in a column matrix.	RSUM	
RTN	Return. In a running program, branches the program pointer back to the line following the most recent XEQ instruction. If there is no matching XEQ instruction, program execution halts. From the keyboard, RTN moves the program pointer to line 00 of the current program.	RTN	
SCI	Selects Scientific display format.	SCI	
SDEV	Standard deviation. Returns s, and s, using the current statistical data.	s	
SEED	Stores a seed for the random number generator.	SEED	
SF	Sets flag nn (00 < nn < 35; 81 < nn < 99).	SF	
SHOW	Shows full precision of the number in the X-register, the entire Alpha register, or a complete program line. (Not programmable)	SHOW	Programmable but don't show anything in programs
SIGN	Sign. Returns 1 for x = 0, -1 for x < 0, and 0 for non-numbers. Returns the unit vector of a complex number.	sign	
SIN	Sine. Returns sin x.	SIN	
SINH	Hyperbolic sine. Returns sinh x.	sinh	
SIZE	Sets the number of storage registers.	N.A.	
SLOPE	Returns the slope of the linear transformation of the current curve-fitting model	L.R.	Returns all parameter of fit curve
SOLVE	Solves for an unknown variable.	SOLVE	
SQRT	Square root. Returns $\sqrt{x}$.	$\sqrt{x}$	
SST	Single step. Moves the program pointer to the next program line. (Not programmable.)	SST	
STO	Stores a copy of x into a destination register or variable.	STO	
STO-	Store subtraction. Subtracts x from an existing register or variable.	STO-	
STO+	Store addition. Adds x to an existing register or variable.	STO+	
STO÷	Store division. Divides an existing register or variable by x.	STO÷	
STOEL	Store element. Stores a copy of x into the current element of the indexed matrix.	STOEL	
STOIJ	Moves the row- and column-pointers to I = y and J = x in	STOIJ	

HP-42S	Description	C47/R47	Comment
	the indexed matrix.		
STOP	Stops program execution.	STOP	
STOx	Store multiplication. Multiplies an existing register or variable by x.	STOx	
STR?	If the X-register contains an Alpha string, executes the next program line; if the X-register does not contain an Alpha string, skips the next program line.	STRI?	
SUM	Returns the sums Σx and Σy into the X- and Y registers.	Σ SUM	
TAN	Tangent. Returns $\tan x$.	TAN	
TANH	Hyperbolic tangent. Returns $\tanh x$.	tanh	
TONE	Sounds a tone.	TONE	
TRACE	Selects Trace printing mode, which prints a record of keystrokes and results.	TRACE	
TRANS	Returns the transpose of the matrix in the X-register.	[M]T	
UVEC	Unit vector. Returns the unit vector for the matrix or complex number in the X-register.	UNITV	
VARMENU	Creates a variable menu using MVAR instructions following the specified global label.	42VRMNU	Same as VARMENU, storing the label of a key pressed without data input in K as the Alpha register
VIEW	Views the contents of a register or variable.	VIEW	
WMEAN	Weighted mean. Returns the mean of x-values weighted by the y-values: $\Sigma xy \div \Sigma y$.	$\bar{x}_w$	
WRAP	Selects Wrap mode, which prevents the indexed matrix from growing.	CF M.GROW	
X<>	Swaps the contents of the X-register with another register or variable.	$x \leftrightarrow$	
X<>Y	Swaps the contents of the X- and Y-registers.	$x \leftrightarrow y$	
X<0?	X less than zero test.	$x < ? 0.$	
X<Y?	X less than y test.	$x < ? Y$	
X=0?	X equal to zero test.	$x = ? 0.$	
X≠0?	X not equal to zero test.	$x \neq ? 0.$	
X=Y?	X equal to y test.	$x = ? Y$	
X≠Y?	X not equal to y test.	$x \neq ? Y$	
X>0?	X greater than zero test.	$x > ? 0.$	
X>Y?	X greater than y test.	$x > ? Y$	
X≤0?	X less than or equal to zero test.	$x \leq ? 0.$	
X≤Y?	X less than or equal to y test.	$x \leq ? Y$	
X≥0?	X greater than or equal to zero test.	$x \geq ? 0.$	
X≥Y?	X greater than or equal to y test.	$x \geq ? Y$	
X ↑ 2	Square. Returns x^2 .	x^2	
XEQ	Execute a function or program.	XEQ	
XOR	Logical XOR (exclusive OR). Returns x XOR y.	XOR	
XTOA	X to Alpha. Appends a character (specified by the code in the X-register) to the Alpha register. If the X-register contains an Alpha string, appends the entire string.	42XTOA	Same as XTOA, using K as the Alpha register
YINT	Y intercept. Returns the y-intercept of the curve fitted to the current statistical data.	L.R.	Returns all parameter of fit curve
Y ↑ X	Power. Returns y^x .	y^x	
1/X	Reciprocal. Returns $1 \div x$.	1/x	
10 ↑ X	Common exponential. Returns 10^x .	10^x	

HP-42S	Description	C47/R47	Comment
Σ^-	Summation minus. Subtracts a pair of x- and y-values from the summation registers.	Σ^-	
Σ^+	Summation plus. Accumulates a pair of x- and y-values into the summation registers.	Σ^+	
+	Addition. Returns $y + x$.	+	
-	Subtraction. Returns $y - x$.	-	
x	Multiplication. Returns $x * y$.	x	
$\div$	Division. Returns $y \div x$.	$\div$	
+/-	Changes the sign of the number in the X-register. While entering an exponent, can also be used to change the sign of the exponent.	CHS	
Σ REG	Summation registers. Defines which storage register begins the block of summation registers.	N.A.	
Σ REG?	Returns the register number of the first summation register.	N.A.	
$\rightarrow$ DEC	To decimal. Converts a real number in octal (base 8) representation to a real number (base 10). Note: This function is included to provide program compatibility with the HP-41 (which uses the function name DEC) and is not related to the Base application		Missing
$\rightarrow$ DEG	To degrees. Converts an angle value in radians to degrees. Returns $x/(2\pi/360)$.	rad $\rightarrow$ deg	rad $\rightarrow$ deg changes a real in X (in CONV:Angle:)
$\rightarrow$ HMS	To hours, minutes, and seconds. Converts x from a decimal fraction to a minutes-seconds format.	HR $\rightarrow$ HMS	In CONV:Misc:
$\rightarrow$ HR	To hours. Converts x from a minutes-seconds format to a decimal fraction.	HMS $\rightarrow$ HR	In CONV:Misc:
$\rightarrow$ OCT	To octal. Converts a real number in decimal (base 10) to a real number in octal representation. Note: This function is included to provide program compatibility with the HP-41 (which uses the function name OCT) and is not related to the Base application		Missing
$\rightarrow$ POL	To polar. Converts x and y to the corresponding polar coordinates r and θ . If the X-register contains a complex number, converts the two parts of the number to polar values.	$\rightarrow$ POL _{hp}	
$\rightarrow$ RAD	To radians. Converts an angle value in degrees to radians. Returns $(2\pi/360)x$.	deg $\rightarrow$ rad	deg $\rightarrow$ rad changes a real in X (in CONV:Angle:)
$\rightarrow$ REC	To rectangular. Converts r (in the X-register) and θ (in the Y-register) to the corresponding rectangular coordinates, x and y. If the X-register contains a complex number, converts the two parts of the number to rectangular values.	$\rightarrow$ RECT _{hp}	
$\leftarrow$	Moves left one element in the indexed matrix.	$\leftarrow$	Not programmable
$\uparrow$	Moves up one element in the indexed matrix.	$\uparrow$	Not programmable
$\downarrow$	Moves down one element in the indexed matrix.	$\downarrow$	Not programmable
$\rightarrow$	Moves right one element in the indexed matrix.	$\rightarrow$	Not programmable
%	Percent. Returns $(x * y) = 100$. (Leaves the y-value in the Y-register.)	%	
%CH	Percent change. Returns $(x - y)(100 \div y)$.	Δ %	

Appendix B: HP-42S to C47/R47 flags mapping

HP-42S			C47/R47	
Type	#	Description	# / Name	Comment
User flag	0-10		0-10	Global user flag
Control flag	11	Automatic Execution	AUTXEQ	
Control flag	12	Double-Wide Printer Output	N.A.	
Control flag	13	Lowercase Printer Output	N.A.	
Control flag	14	Reserved		
Control flag	15	Print Mode.	TRACE	
Control flag	16		NORM	
Control flag	17	Reserved		
Control flag	18			
Control flag	19	General Use	19	Global user flag
Control flag	20		20	Global user flag
Control flag	21	Printer Enable	EN	
Control flag	22	Numeric Data Input	NUM.IN	
Control flag	23	Alpha Data Input	ALP.IN	
Control flag	24	Range Error Ignore	N.A.	
Control flag	25	Error Ignore	IGN1ER	
Control flag	26	Audio Enable	QUIET	Inverse logic
Control flag	27	CUSTOM Menu	N.A.	
Control flag	28	Radix Mark (. or ,)	N.A.	
Control flag	29	Digit Separators	N.A.	
Control flag	30	Stack Lift Disable	ASLIFT	Inverse logic
Control flag	31	Reserved		
Control flag	32			
Control flag	33			
Control flag	34	AGRAPH Control		Reserved variable GRAMOD
Control flag	35			
System flag	36	Number of Digits		NDEC# returns the number of digits
System flag	37			
System flag	38			
System flag	39			
System flag	40	Display Format		DF# returns the display format
System flag	41			
System flag	42	Grads Mode		ADM# returns the current Angle Mode
System flag	43	Radians Mode		
System flag	44	Continuous On	AUTOFF	Inverse logic
System flag	45	Solving	SOLVING	
System flag	46	Integrating	INTING	

HP-42S			C47/R47	
System flag	47	Variable Menu	VMDISP	
System flag	48	Alpha Mode	ALPHA	
System flag	49	Low Battery	LOWBAT	
System flag	50	Message	N.A.	
System flag	51	Two-Line Message	N.A.	
System flag	52	Program-Entry	N.A.	
System flag	53	INPUT	N.A.	
System flag	54	Reserved		
System flag	55	Printer Existence	ACT	
System flag	56	Linear Curve-Fitting Model		BestF# returns the Fitting models selected
System flag	57	Logarithmic Curve-Fitting Model		
System flag	58	Exponential Curve-Fitting Model		
System flag	59	Power Curve-Fitting Model		
System flag	60	AllΣ Mode (statistics)	N.A.	
System flag	61	Logarithmic Model Invalid	N.A.	
System flag	62	Exponential Model Invalid	N.A.	
System flag	63	Power Model Invalid Invalid	N.A.	
System flag	64	Reserved		
System flag	65	Matrix Editor	N.A.	
System flag	66	Grow Mode	GROW	
System flag	67	42S: Reserved – Free4: DATE mode with flag 31		
System flag	68	Base Mode	N.A.	
System flag	69			
System flag	70			
System flag	71			
System flag	72	Local Label Mode (CUSTOM)	N.A.	
System flag	73	Polar Mode	POLAR	
System flag	74	Real-Result Only	CPXRES	Inverse logic
System flag	75	Programmable Menu Selected	N.A.	
System flag	76	Edge Wrap	WRPEDG	
System flag	77	End Wrap	WRPEND	
System flag	78	42S: Reserved – Free4: BSIGNED mode for BASE		ISM/ISM# sets/gets integer sign mode
System flag	79	42S: Reserved – Free4: BWRAP mode for BASE		
System flag	80	42S: Reserved – Free4: Big Stack mode		
User flag	81-99		81-99	Global user flag

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